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Experiment:- 04

AIM:-Numerical Computation with NumPy: Perform different computations using NumPy library.

Q1. Explain the role of NumPy in the Python scientific computing ecosystem.

ANS:- NumPy (Numerical Python) is the core library for scientific computing in Python. It provides support for large, multi-dimensional arrays and matrices, along with a wide range of mathematical functions to operate on them. Its key roles include: Acting as the foundation for many other scientific libraries. Enabling efficient storage and manipulation of numerical data Supporting vectorized operations, reducing the need for loops. Serving as the base for libraries like Pandas, Matplotlib, and Scikit-learn.

```
import numpy as np
arr = np.array([1, 2, 3, 4, 5])
print("NumPy Array:", arr)
print("Array Type:", type(arr))
```

```
NumPy Array: [1 2 3 4 5]
Array Type: <class 'numpy.ndarray'>
```

Q2. Why is NumPy considered faster than pure Python computations?

ANS:- NumPy is faster than pure Python because: Vectorization: Operations are applied to entire arrays at once instead of element-by-element loops Optimized C Implementation: NumPy functions are wriflen in C, making them highly efficient. Less Overhead: Python lists store mixed data types, while NumPy arrays store homogeneous data, reducing memory and computation overhead Befler Memory Usage: Data is stored in contiguous memory blocks, improving access speed

```
import numpy as np
import time
lst = list(range(1000000))
start = time.time()
lst = [x * 2 for x in lst]
print("Python Time:", time.time() - start)
arr = np.arange(1000000)
start = time.time()
```

```
arr = arr * 2
print("NumPy Time:", time.time() - start)
```

```
Python Time: 0.07217717170715332
NumPy Time: 0.0041656494140625
```

Q3.What types of mathematical operations can be performed using NumPy?

ANS:- NumPy supports a wide range of operations:

Arithmetic operations: +, −, ×, ÷

Statistical operations: mean, median, standard deviation.

Linear algebra: dot product, matrix multiplication.

Trigonometric functions: sin, cos, tan.

Logarithmic & exponential functions

```
import numpy as np
arr = np.array([1, 2, 3, 4])
print("Addition:", arr + 2)
print("Mean:", np.mean(arr))
print("Sine:", np.sin(arr))
print("Exponential:", np.exp(arr))
```

```
Addition: [3 4 5 6]
Mean: 2.5
Sine: [ 0.84147098  0.90929743  0.14112001 -0.7568025 ]
Exponential: [ 2.71828183  7.3890561  20.08553692 54.59815003]
```

Q4. Explain indexing and slicing in NumPy arrays.

ANS:-. Indexing and slicing allow accessing specific elements or subsets of arrays.

Indexing: Access single elements

Slicing: Extract portions of arrays

Example:-

arr[0] → first element arr[1:4] → elements from index 1 to 3

NumPy also supports: Multi-dimensional indexing

Boolean indexing

```
import numpy as np
arr = np.array([10, 20, 30, 40, 50])
print("First Element:", arr[0])
print("Elements from index 1 to 3:", arr[1:4])
```

```
First Element: 10
Elements from index 1 to 3: [20 30 40]
```

Q5. How does NumPy integrate with libraries like:

ANS:- NumPy integrates seamlessly with other Python libraries, making it a core component of the data science ecosystem.

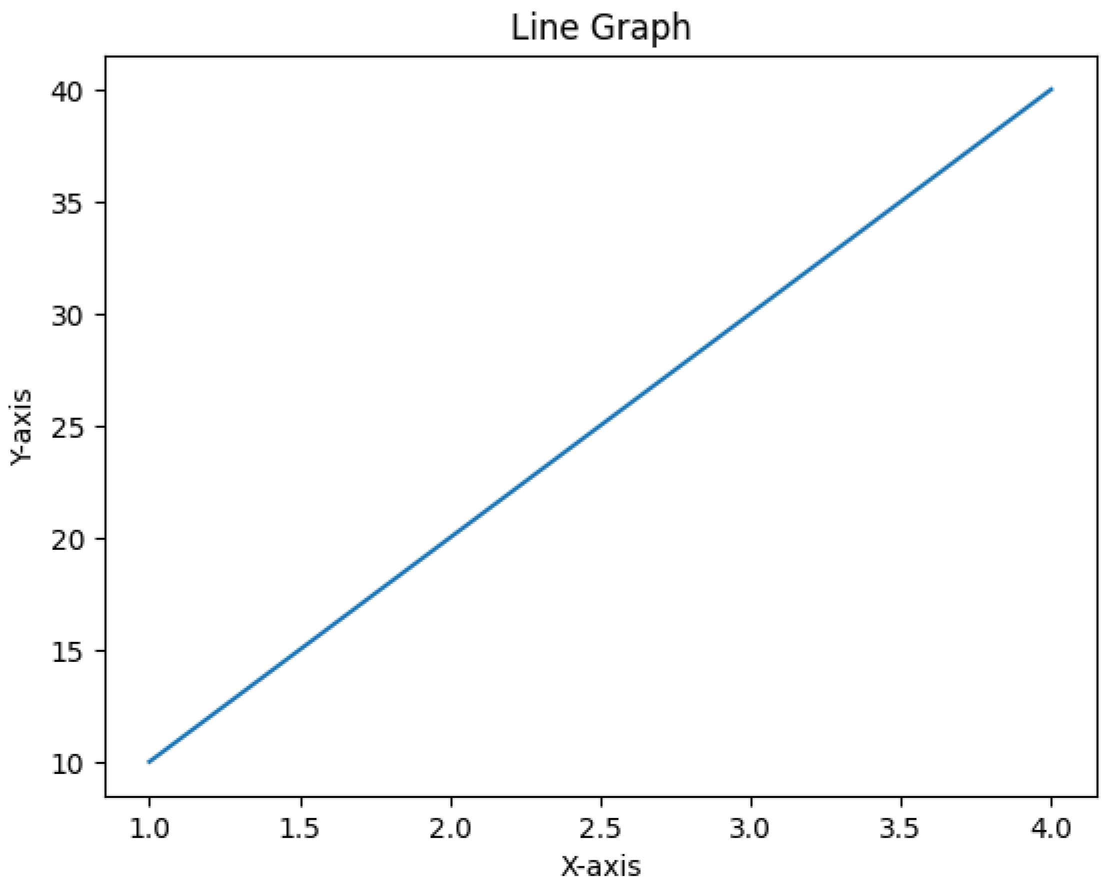
A) Pandas:- Pandas is a data analysis library that provides data structures like DataFrames and Series. It uses NumPy arrays internally for fast data processing and manipulation.

```
import numpy as np
import pandas as pd
data = np.array([10, 20, 30, 40])
df = pd.DataFrame(data, columns=["Values"])
print(df)
```

```
   Values
0      10
1      20
2      30
3      40
```

B) Matplotlib:- Matplotlib is a data visualization library used to create graphs, charts, and plots. It uses NumPy arrays to represent numerical data for plofling.

```
import numpy as np
import matplotlib.pyplot as plt
x = np.array([1, 2, 3, 4])
y = np.array([10, 20, 30, 40])
plt.plot(x, y)
plt.title("Line Graph")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.show()
```



C) Scikit-learn:- Scikit-learn is a machine learning library used for building models like regression, classification, and clustering. It takes NumPy arrays as input for training and prediction.

```
import numpy as np
from sklearn.linear_model import LinearRegression
X = np.array([[1], [2], [3], [4]])
y = np.array([10, 20, 30, 40])
model = LinearRegression()
model.fit(X, y)
print("Prediction for 5:", model.predict([[5]]))
```

```
Prediction for 5: [50.]
```

Conclusion: The practical on Numerical Computation with NumPy demonstrates how efficiently numerical operations can be performed using array-based computation. It highlights NumPy's ability to handle large datasets, perform mathematical and statistical operations, and execute calculations faster than traditional methods. Overall, the experiment shows that NumPy is a powerful and essential library for scientific computing, making numerical tasks simpler, faster, and more efficient.